CS23A34
USER INTERFACE DESIGN
EXPERIMENT-2

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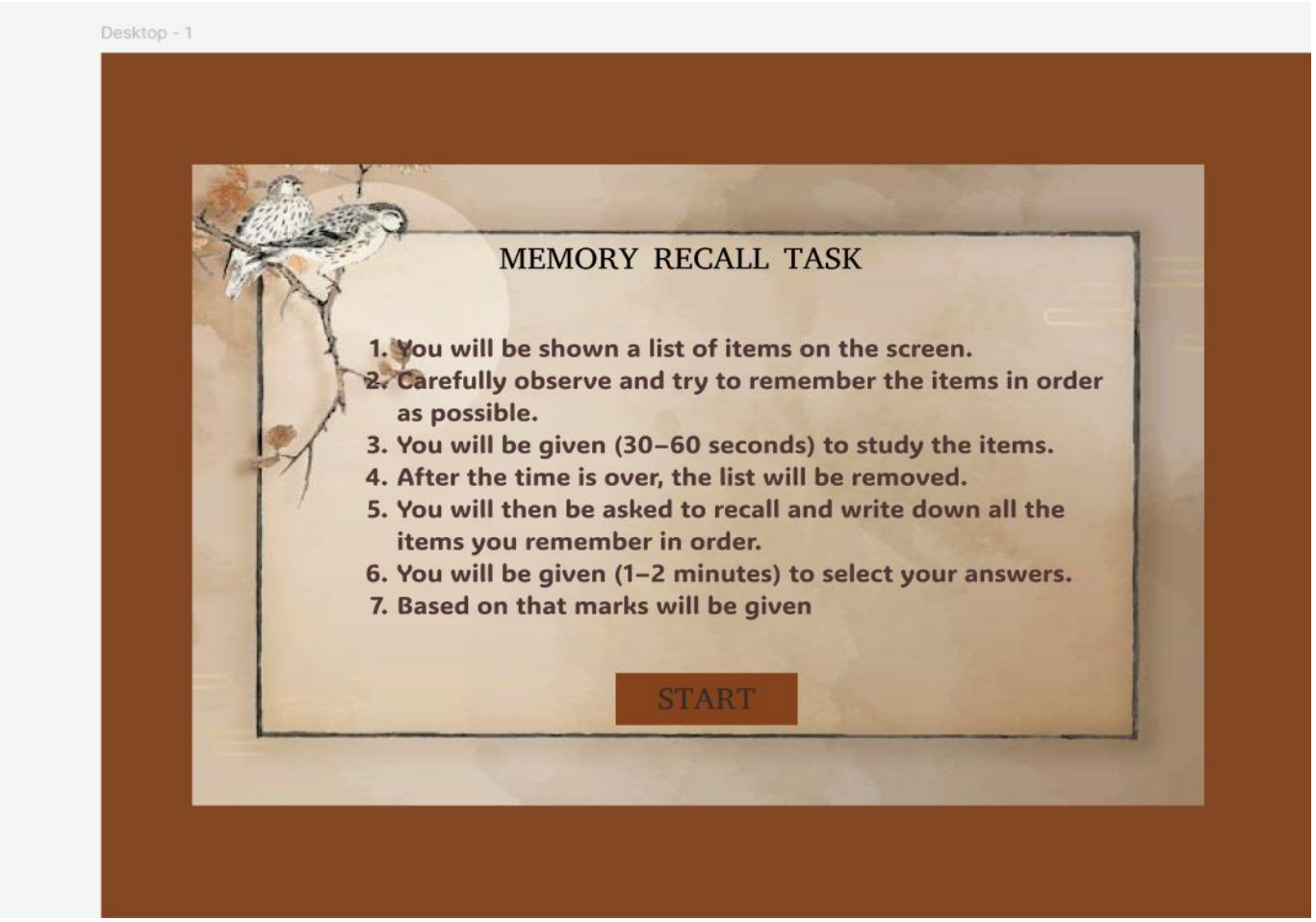
Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on user memory.

FRAME 1: INSTRUCTION PAGE



Chunking Analysis of the Instruction Page

Chunking is a cognitive technique that helps users understand and remember information by breaking it into smaller, organized units. The instruction page of the **Memory Builder** application effectively applies chunking principles to prepare users for the task.

1. Step-by-Step Information Breakdown

- The instructions are presented in a **numbered sequence**, guiding users from start to finish.
- Each step focuses on **one specific action**, making the task easy to understand.
- This structured flow reduces confusion and mental overload.

2. Logical Flow of Task Stages

- Initial steps explain **category selection** and **viewing items**.
- Middle steps focus on **memorization and arrangement**.
- Final steps inform users about **submission and score display**.
- This ordered grouping helps users mentally prepare for each phase.

3. Visual and Textual Chunking

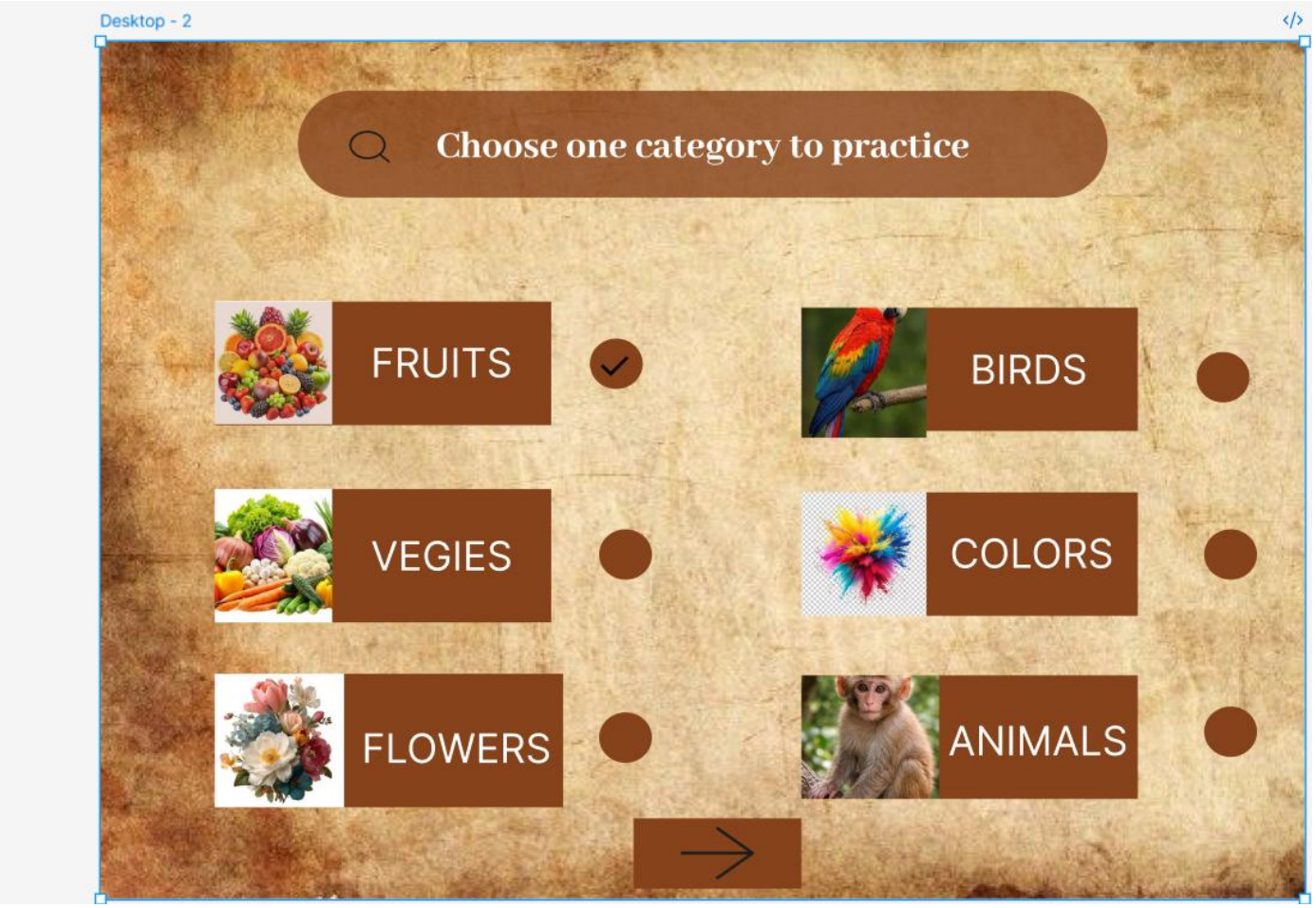
- The bold title **“MEMORY BUILDER”** clearly defines the purpose of the page.

- Instructions are grouped together in one section, separated from actions.
- Adequate spacing improves readability and focus.

4. Clear Call-to-Action

- The **“START”** button is visually separated and placed at the bottom.
- This clearly indicates the next step without overwhelming the user.

FRAME 2: CATEGORY SELECTION PAGE



Chunking Analysis of the Category Selection Page

This frame allows users to select a category before starting the memory task. The design applies chunking by organizing information into meaningful groups.

1. Categorized Information Structure

- Items are divided into clear categories such as **Fruits, Games, Flowers, Animals, Birds, and Colours**.
- Each category represents a single chunk, making selection easier.

2. Visual Chunking Using Images

- Each category is represented using images along with text labels.
- Visual cues help users quickly recognize and remember their chosen category.

3. Reduced Decision Load

- Users select only **one category at a time**, limiting mental effort.
- The clean layout avoids unnecessary distractions.

4. Guided Navigation

- The **"NEXT"** button clearly indicates progression.
- This ensures users understand the flow of the task.

FRAME 3: REMEMBER ITEMS IN ORDER PAGE

Chunking Analysis of the Memorization Page

This frame demonstrates the core idea of chunking by dividing items into small, manageable groups for memorization.

1. Group-Based Chunking

- Items are divided into **Group 1, Group 2, and Group 3**.
- Each group contains a limited number of items, improving memory retention.

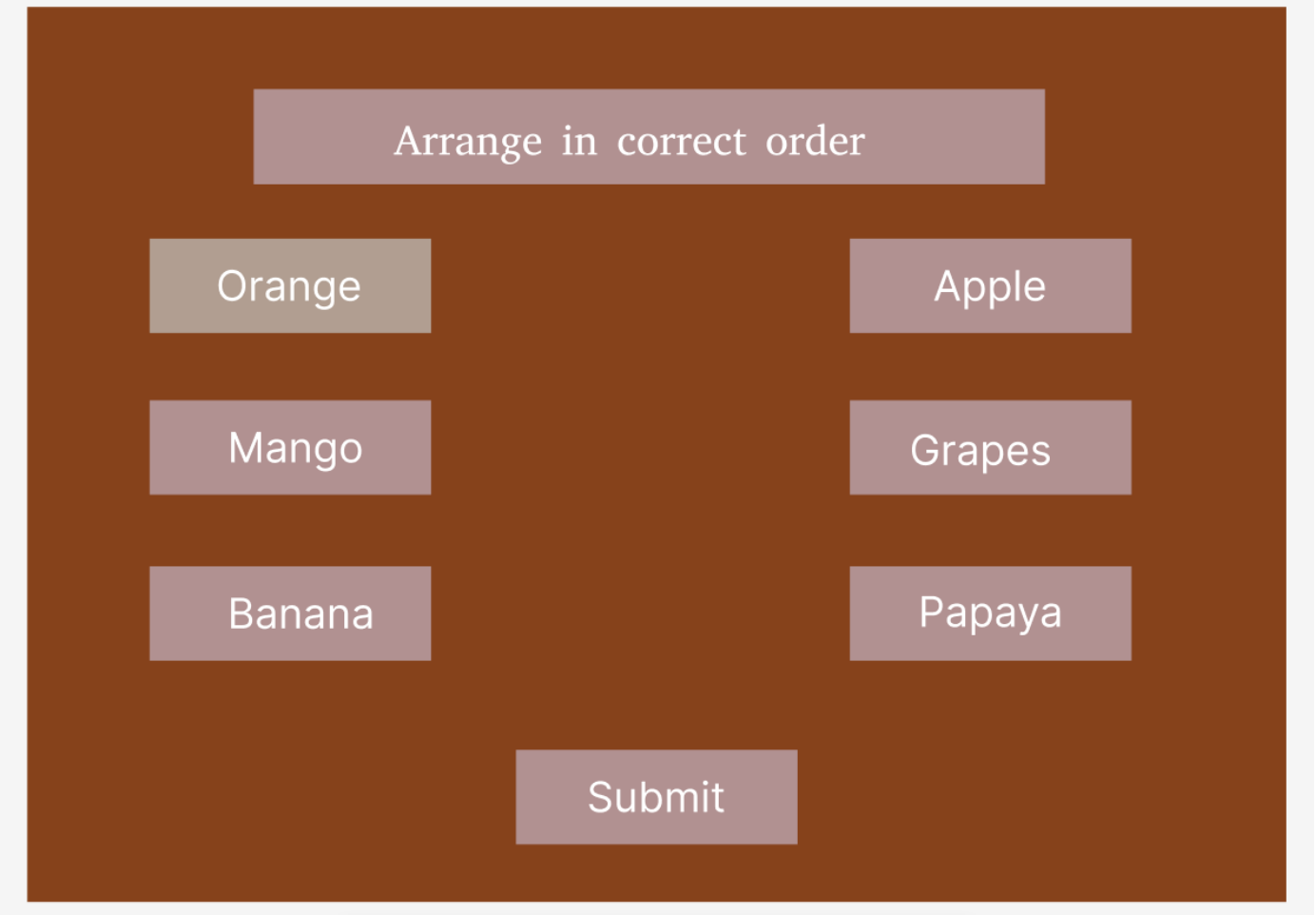
2. Sequential Presentation

- Items are shown in a fixed order, encouraging structured memorization.
- Clear group labels help users mentally separate information.

3. Focused Display

- Only memorization-related content is shown on the screen.
- This minimizes distractions and improves concentration. **4. Clear Transition to Recall**
- The **"Start Test"** button signals the end of memorization and the start of recall.

FRAME 4: ARRANGE IN CORRECT ORDER PAGE



Chunking Analysis of the Recall Page

This frame evaluates how effectively users recall the previously chunked information.

1. Individual Item Presentation

- Each item is shown separately, allowing users to compare and arrange them.
- This visual separation supports active recall.

2. Task-Focused Design

- The heading **"Arrange in correct order"** clearly states the task.
- No additional information distracts the user.

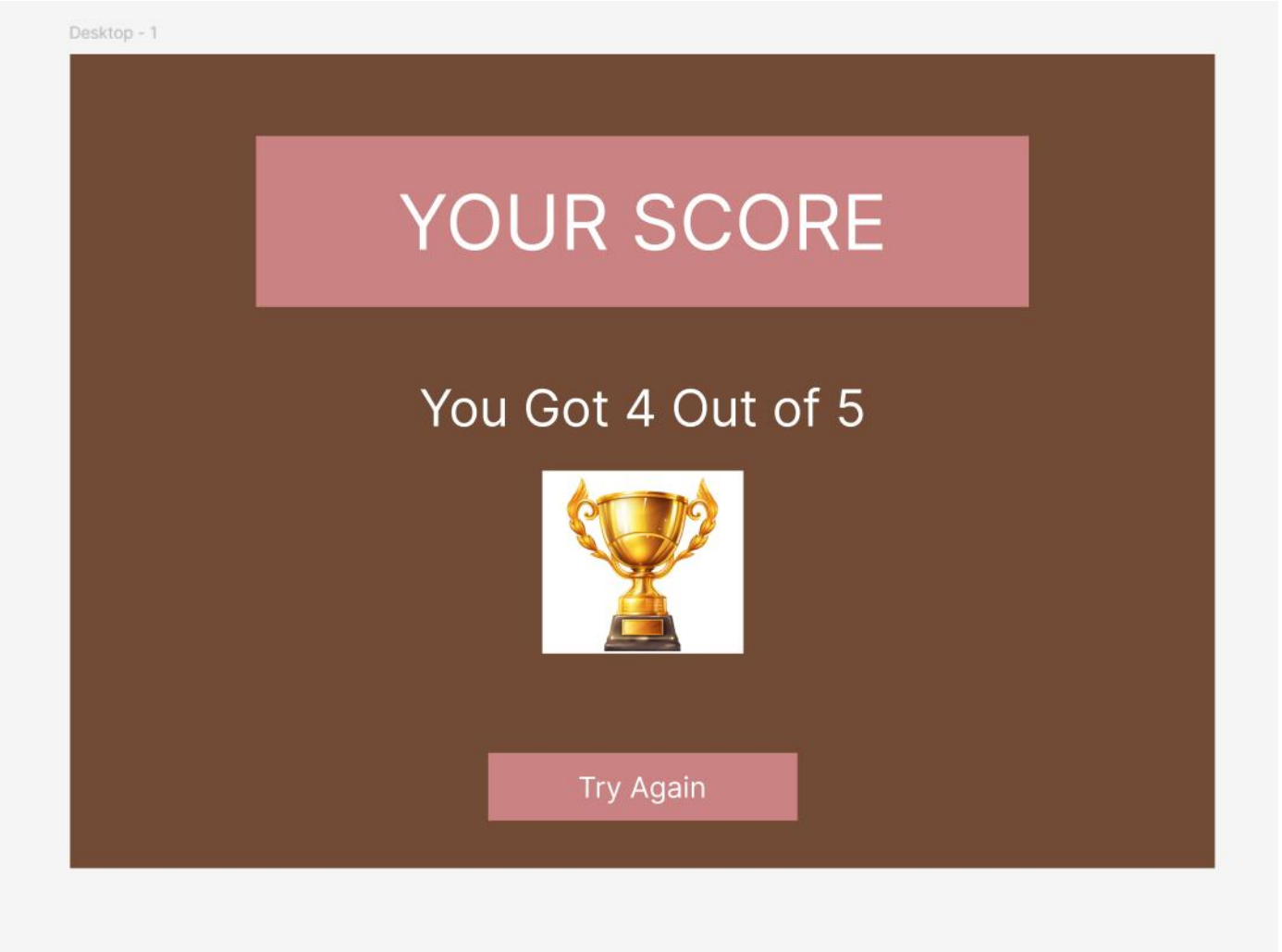
3. Cognitive Reinforcement

- Rearranging items reinforces memory by engaging users actively.

4. Clear Completion Action

- The **“Submit”** button finalizes the recall phase.

FRAME 5: SCORE PAGE



Chunking Analysis of the Result Page

The score page summarizes performance in a simple and motivating manner. **1. Simplified Feedback**

- The result is shown as **“You got X out of Y correct”**, converting performance into a single chunk.

2. Visual Reinforcement

- Emojis and trophy icons provide positive feedback.
- This enhances user motivation.

3. Minimal Information Load

- Only essential information is displayed, avoiding clutter.

4. Encouragement for Reattempt

- The **“Try Again”** button allows users to repeat the task and improve memory.